CI/CD pipelines

Tier: Free, Premium, Ultimate

Offering: GitLab.com, GitLab Self-Managed, GitLab Dedicated

CI/CD pipelines are the fundamental component of GitLab CI/CD. Pipelines are configured in a `.gitlab-ci.yml` file by using YAML keywords.

Pipelines can run automatically for specific events, like when pushing to a branch, creating a merge request, or on a schedule. When needed, you can also run pipelines manually.

Pipelines are composed of:

- Global YAML keywords that control the overall behavior of the project's pipelines.
- Jobs that execute commands to accomplish a task. For example, a job could compile, test, or deploy code. Jobs run independently from each other, and are executed by runners.
- Stages, which define how to group jobs together. Stages run in sequence, while the jobs in a stage run in parallel. For example, an early stage could have jobs that lint and compile code, while later stages could have jobs that test and deploy code. If all jobs in a stage succeed, the pipeline moves on to the next stage. If any job in a stage fails, the next stage is not (usually) executed and the pipeline ends early.

A small pipeline could consist of three stages, executed in the following order:

- A `build` stage, with a job called `compile` that compiles the project's code.
- A `test` stage, with two jobs called `test1` and `test2` that run various tests on the code. These tests would only run if the `compile` job completed successfully.
- A `deploy` stage, with a job called `deploy-to-production`. This job would only run if both jobs in the `test` stage started and completed successfully.

To get started with your first pipeline, see Create and run your first GitLab CI/CD pipeline.

Types of pipelines

Pipelines can be configured in many different ways:

- Basic pipelines run everything in each stage concurrently, followed by the next stage.
- Pipelines that use the `needs` keyword run based on dependencies between jobs and can run more quickly than basic pipelines.
- Merge request pipelines run for merge requests only (rather than for every commit).
- Merged results pipelines are merge request pipelines that act as though the changes from the source branch have already been merged into the target branch.
- Merge trains use merged results pipelines to queue merges one after the other.
- Workload pipelines run on ephemeral Git references for on-demand pipeline execution without creating temporary branches.
- Parent-child pipelines break down complex pipelines into one parent pipeline that can trigger multiple child sub-pipelines, which all run in the same project and with the same SHA. This pipeline architecture is commonly used for mono-repos.
- Multi-project pipelines combine pipelines for different projects together.

Pipelines and their component jobs and stages are defined with YAML keywords in the CI/CD pipeline configuration file for each project. When editing CI/CD configuration in GitLab, you should use the pipeline editor.

You can also configure specific aspects of your pipelines through the GitLab UI:

- Pipeline settings for each project.
- Pipeline schedules.
- Custom CI/CD variables.

If you use VS Code to edit your GitLab CI/CD configuration, the GitLab for VS Code extension helps you validate your configuration and view your pipeline status.

Run a pipeline manually

History >

Pipelines can be manually executed, with predefined or manually-specified variables.

You might do this if the results of a pipeline (for example, a code build) are required outside the standard operation of the pipeline.

To execute a pipeline manually:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Build** > **Pipelines**.
3. Select **New pipeline**.
4. In the **Run for branch name or tag** field, select the branch or tag to run the pipeline for.
5. Optional. Enter any:
   - Inputs required for the pipeline to run. Default values for inputs are prefilled, but can be modified. Input values must follow the expected type.
   - CI/CD variables. You can configure variables to have their values prefilled in the form. Using inputs to control pipeline behavior offers improved security and flexibility over CI/CD variables.
6. Select **New pipeline**.

The pipeline now executes the jobs as configured.

View manual pipeline variables

History >

You can see all variables that are specified when the pipeline runs manually.

Prerequisites:

- You must have the Owner role for the project.

The required role depends on what you want to do:

Action	Minimum role
View variable names	Guest
View variable values	Developer
Configure visibility setting	Owner

external secrets management instead of manual pipeline variables.

To view manual pipeline variables:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Settings** > **CI/CD**.
3. Select **Display pipeline variables**.
4. Go to **Build** > **Pipelines** and select a pipeline that was run manually.
5. Select the **Manual Variables** tab.

Variable values are masked by default. If you have the Developer, Maintainer, or Owner role, you can select the eye icon to reveal values.

Prefill variables in manual pipelines

History >

You can use the `description` and `value` keywords to define pipeline-level (global) variables that are prefilled when running a pipeline manually. Use the description to explain information such as what the variable is used for, and what the acceptable values are. You can use Markdown in the description.

Job-level variables cannot be pre-filled.

In manually-triggered pipelines, the **New pipeline** page displays all pipeline-level variables that have a `description` defined in the `.gitlab-ci.yml` file. The description displays below the variable.

You can change the prefilled value, which overrides the value for that single pipeline run. Any variables overridden by using this process are expanded and not masked. If you do not define a `value` for the variable in the configuration file, the variable name is still listed, but the value field is blank.

For example:

```yaml
variables:
  DEPLOY_CREDENTIALS:
    description: "The deployment credentials."
  DEPLOY_ENVIRONMENT:
    description: "Select the deployment target. Valid options are: 'canary', 'staging', 'production', or a stable branch of your choice."
    value: "canary"
```

In this example:

- `DEPLOY_CREDENTIALS` is listed in the **New pipeline** page, but with no value set. The user is expected to define the value each time the pipeline is run manually.
- `DEPLOY_ENVIRONMENT` is pre-filled in the **New pipeline** page with `canary` as the default value, and the message explains the other options.

> Because of a known issue, projects that use compliance pipelines can have prefilled variables not appear when running a pipeline manually. To workaround this issue, change the compliance pipeline configuration.

Configure a list of selectable prefilled variable values

History >

string in `value` must also be included in the `options` list.

For example:

```yaml
variables:
  DEPLOY_ENVIRONMENT:
    value: "staging"
    options:
      - "production"
      - "staging"
      - "canary"
    description: "The deployment target. Set to 'staging' by default."
```

Run a pipeline by using a URL query string

You can use a query string to pre-populate the **New pipeline** page. For example, the query string `.../pipelines/new?ref=my_branch&var[foo]=bar&file_var[file_foo]=file_bar` pre-populates the **New pipeline** page with:

- **Run for** field: `my_branch`.
- **Variables** section:
  - Variable:
    - Key: `foo`
    - Value: `bar`
  - File:
    - Key: `file_foo`
    - Value: `file_bar`

The format of the `pipelines/new` URL is:

```
.../pipelines/new?ref=<branch>&var[<variable_key>]=<value>&file_var[<file_key>]=<value>
```

The following parameters are supported:

- `ref`: specify the branch to populate the **Run for** field with.
- `var`: specify a `Variable` variable.
- `file_var`: specify a `File` variable.

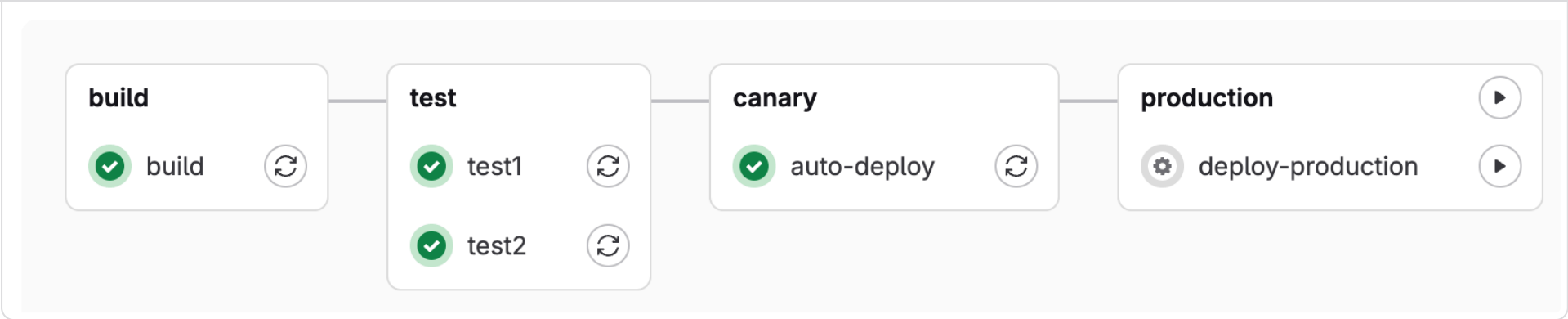
For each `var` or `file_var`, a key and value are required.

Add manual interaction to your pipeline

Manual jobs, allow you to require manual interaction before moving forward in the pipeline.

You can do this straight from the pipeline graph. Select **Run** (▶) to execute that particular job.

For example, your pipeline can start automatically, but require a manual action to deploy to production. In the following example, the `production` stage has a job with a manual action:



Start all manual jobs in a stage

If a stage contains only manual jobs, you can start all the jobs at the same time by selecting **Run all manual** (▶) above the stage. If the stage contains non-manual jobs, the option is not displayed.

Skip a pipeline

To push a commit without triggering a pipeline, add `[ci skip]` or `[skip ci]`, using any capitalization, to your commit message.

Alternatively, with Git 2.10 or later, use the `ci.skip` Git push option. The `ci.skip` push option does not skip merge request pipelines.

When you skip a pipeline:

- An empty pipeline, with no jobs or stages, is still created in GitLab. The pipeline appears in the UI and can be returned in API responses.
- The pipeline status is **Skipped** in the UI, and `skipped` in the API.

> Pipeline execution policies and scan execution policies can restrict or disable the `[skip ci]` directive. For more information, see:
>
> - The `skip_ci` type in pipeline execution policies.
> - The `skip_ci` type in scan execution policies.

Delete a pipeline

Users with the Owner role for a project can delete a pipeline:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Build** > **Pipelines**.
3. Select either the pipeline ID (for example `#123456789`) or the pipeline status icon (for example **Passed**) of the pipeline to delete.
4. In the upper right of the pipeline details page, select **Delete**.

Deleting a pipeline does not automatically delete its child pipelines. See issue 39503 for more details.

> Deleting a pipeline expires all pipeline caches, and deletes all immediately related objects, such as jobs, logs, artifacts, and triggers. **This action cannot be undone**.

Pipeline security on protected branches

A strict security model is enforced when pipelines are executed on protected branches.

The following actions are allowed on protected branches if the user is allowed to merge or push to that specific branch:

- Run manual pipelines (using the Web UI or pipelines API).

- Run on-demand DAST scan.
- Run manual jobs on existing pipelines.
- Retry or cancel existing jobs (using the Web UI or pipelines API).

Variables marked as **protected** are accessible to jobs that run in pipelines for protected branches. Only assign users the right to merge to protected branches if they have permission to access sensitive information like deployment credentials and tokens.

Runners marked as **protected** can run jobs only on protected branches, preventing untrusted code from executing on the protected runner and preserving deployment keys and other credentials from being unintentionally accessed. To ensure that jobs intended to be executed on protected runners do not use regular runners, they must be tagged accordingly.

Review how access to protected variables and runners work in the context of Merge request pipelines.

Review the deployment safety page for additional security recommendations for securing your pipelines.

Trigger a pipeline when an upstream project is rebuilt

Tier: Premium, Ultimate

Offering: GitLab.com, GitLab Self-Managed, GitLab Dedicated

You can set up your project to automatically trigger a pipeline based on tags in a different project. When a new tag pipeline in the subscribed project finishes, it triggers a pipeline on your project's default branch, regardless of the tag pipeline's success, failure, or cancellation.

As an alternative, you can use CI/CD jobs with pipeline trigger tokens to trigger pipelines when another pipeline runs. This method is more reliable and flexible than pipeline subscriptions and is the recommended approach.

Prerequisites:

- The upstream project must be public.
- The user must have the Developer role in the upstream project.

To trigger the pipeline when the upstream project is rebuilt:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Settings** > **CI/CD**.
3. Expand **Pipeline subscriptions**.
4. Select **Add project**.
5. Enter the project you want to subscribe to, in the format `<namespace>/<project>`. For example, if the project is `https://gitlab.com/gitlab-org/gitlab`, use `gitlab-org/gitlab`.
6. Select **Subscribe**.

The maximum number of upstream pipeline subscriptions is 2 by default, for both the upstream and downstream projects. On GitLab Self-Managed, an administrator can change this limit.

How pipeline duration is calculated

The total running time for a given pipeline excludes:

- The duration of the initial run for any job that is retried or manually re-run.
- Any pending (queue) time.

That means that if a job is retried or manually re-run, only the duration of the latest run is included in the total running time.

Each job is represented as a `Period`, which consists of:

A simple example is:

- A (0, 2)
- A' (2, 4)
  - This is retrying A
- B (1, 3)
- C (6, 7)

In the example:

- A begins at 0 and ends at 2.
- A' begins at 2 and ends at 4.
- B begins at 1 and ends at 3.
- C begins at 6 and ends at 7.

Visually, it can be viewed as:

```
0  1  2  3  4  5  6  7
AAAAAAA
   BBBBBBB
      A'A'A'A
                  CCCC
```

Because A is retried, it is ignored, and only job A' is counted. The union of B, A', and C is (1, 4) and (6, 7). Therefore, the total running time is:

```
(4 - 1) + (7 - 6) => 4
```

View pipelines

To view all the pipelines that ran for your project:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Build** > **Pipelines**.

You can filter the **Pipelines** page by:

- Trigger author
- Branch name
- Status
- Tag
- Source

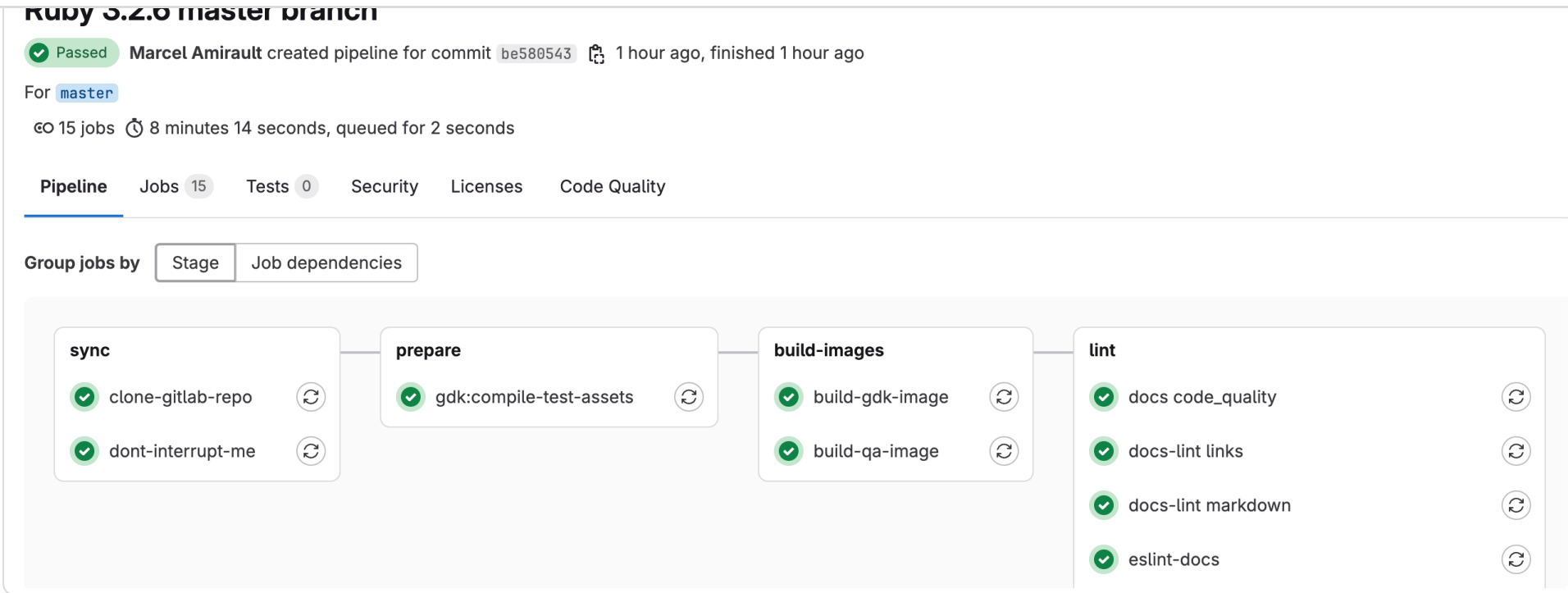
Select **Pipeline ID** in the dropdown list in the upper right to display the pipeline IDs (unique ID across the instance). Select **pipeline IID** to display the pipeline IIDs (internal ID, unique across the project only).

To view the pipelines that relate to a specific merge request, go to the **Pipelines** tab in the merge request.

Pipeline details

History >

Select a pipeline to open the pipeline details page which shows every job in the pipeline. From this page you can cancel a running pipeline, retry failed jobs, or delete a pipeline.

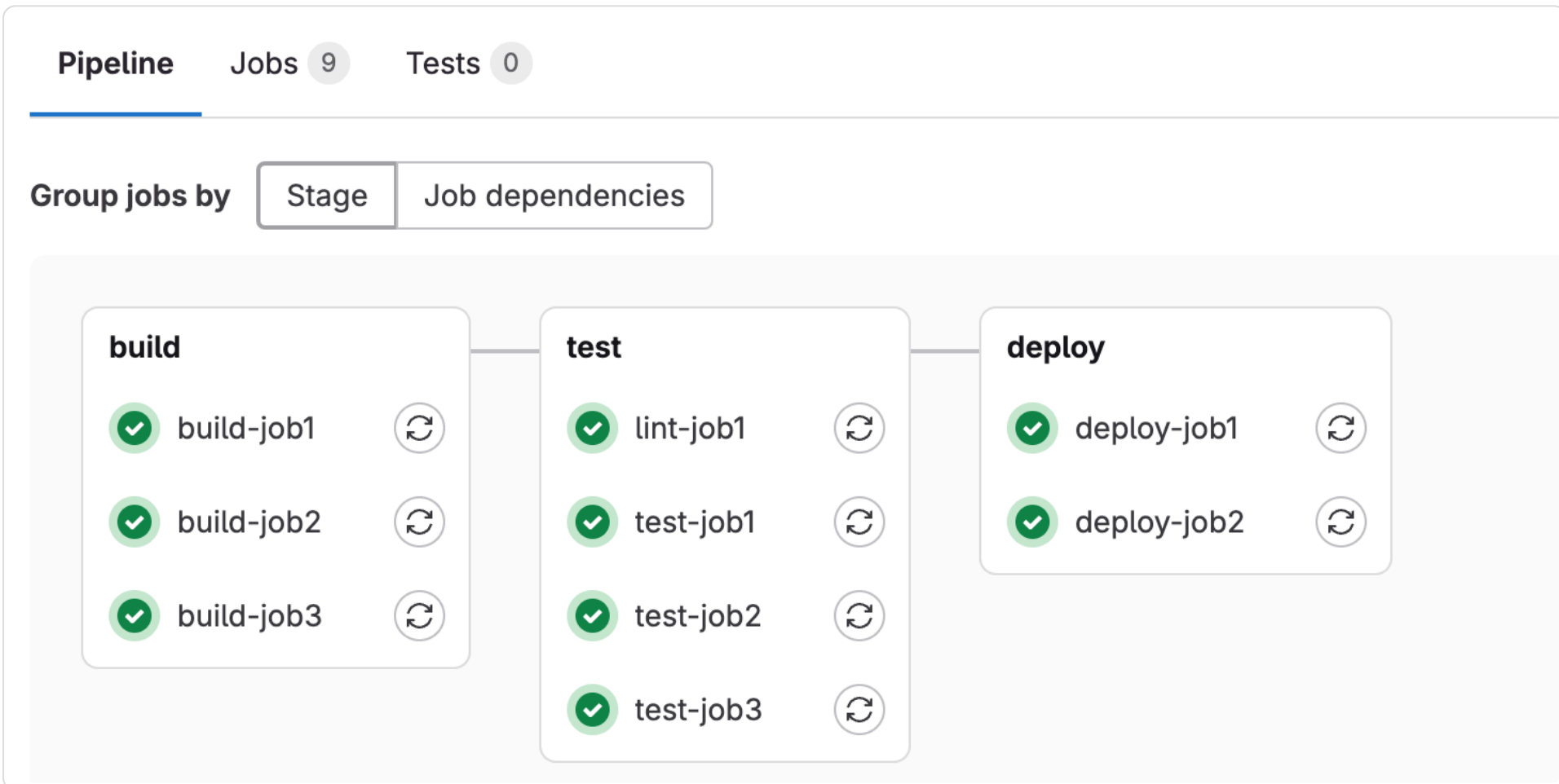


You can use a standard URL to access the details for specific pipelines:

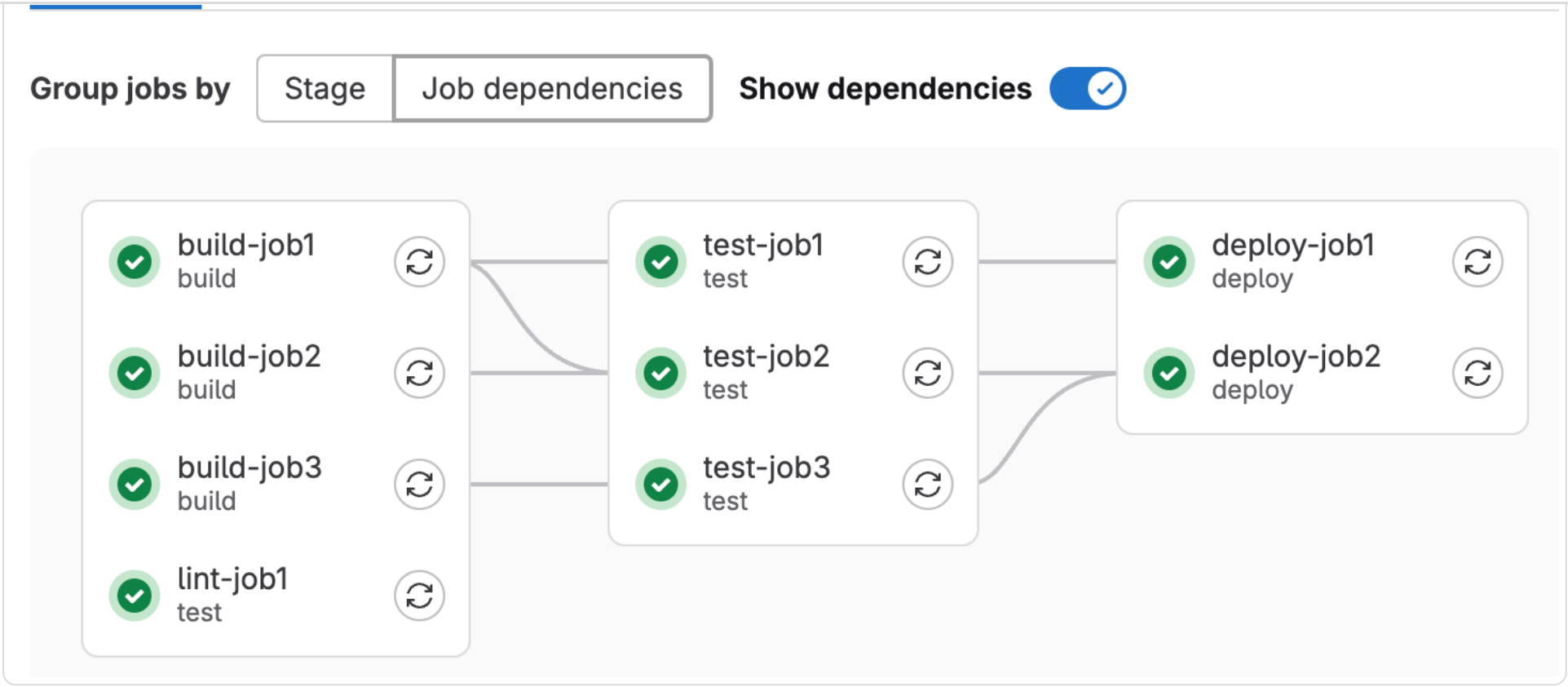
- `gitlab.example.com/my-group/my-project/-/pipelines/latest`: The details page for the latest pipeline for the most recent commit on the default branch in the project.
- `gitlab.example.com/my-group/my-project/-/pipelines/<branch>/latest`: The details page for the latest pipeline for the most recent commit on branch `<branch>` in the project.

Group jobs by stage or `needs` configuration

When you configure jobs with the `needs` keyword, you have two options for how to group the jobs in the pipeline details page. To group the jobs by stage configuration, select **stage** in the **Group jobs by** section:



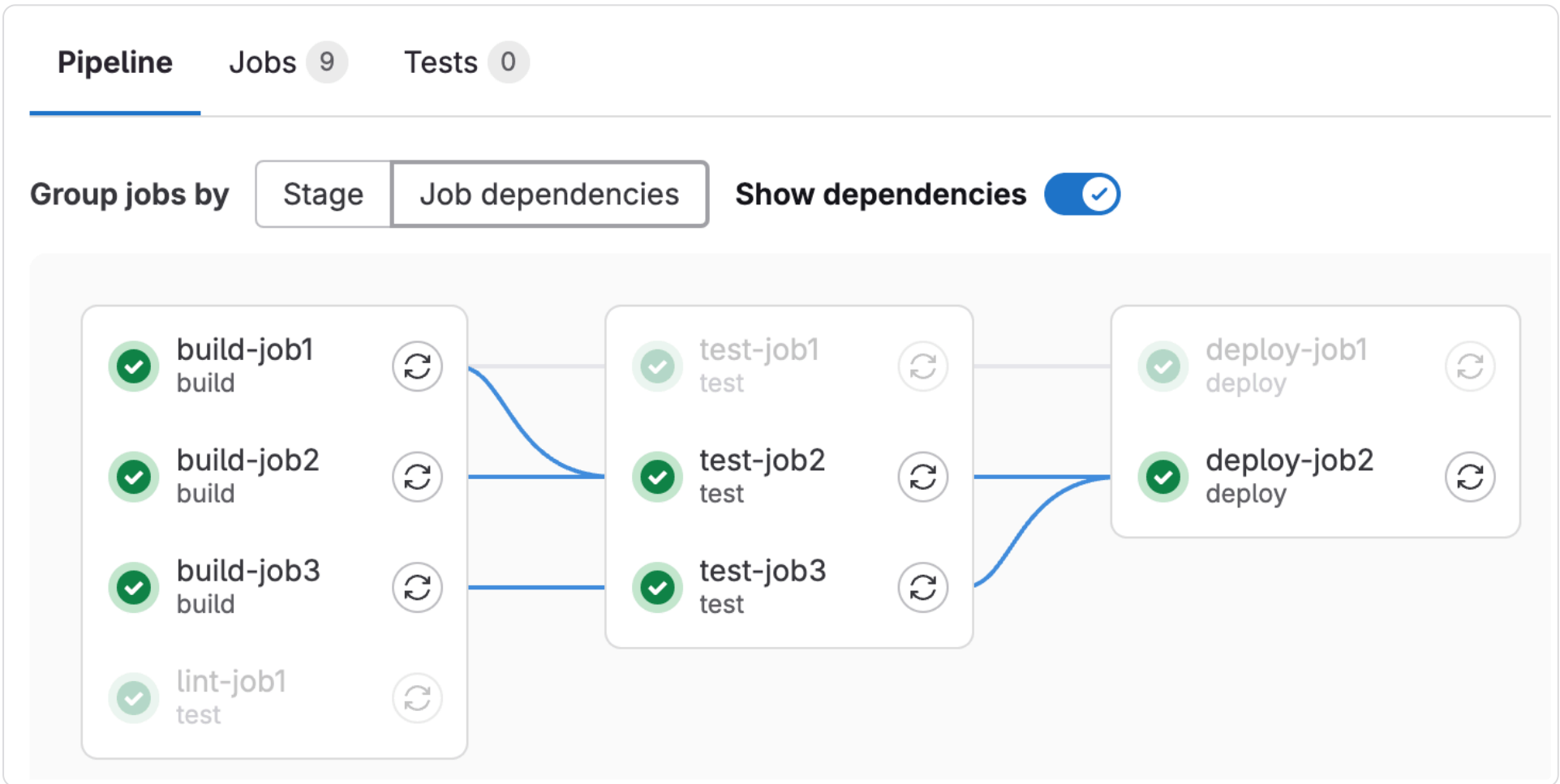
To group the jobs by `needs` configuration, select **Job dependencies**. You can optionally select **Show dependencies** to render lines between dependent jobs.



Jobs in the leftmost column run first, and jobs that depend on them are grouped in the next columns. In this example:

- `lint-job` is configured with `needs: []` and depends on no jobs, so it displays in the first column, despite being in the `test` stage.
- `test-job1` depends on `build-job1`, and `test-job2` depends on both `build-job1` and `build-job2`, so both test jobs display in the second column.
- Both `deploy` jobs depend on jobs in second column (which themselves depend on other earlier jobs), so the deploy jobs display in the third column.

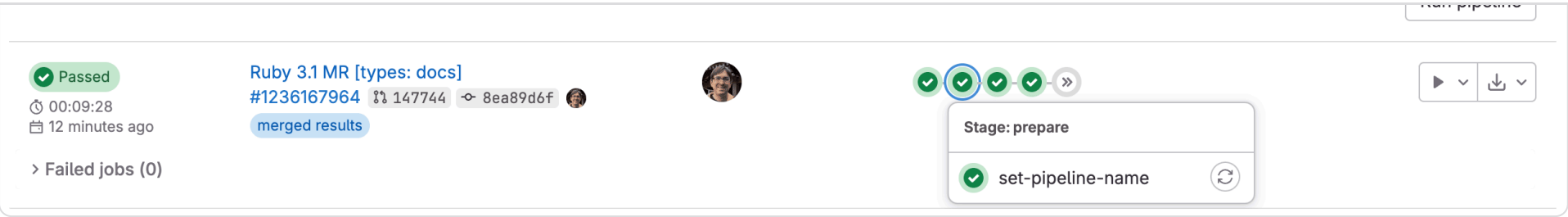
When you hover over a job in the **Job dependencies** view, every job that must run before the selected job is highlighted:



Pipeline mini graphs

Pipeline mini graphs take less space and can tell you at a quick glance if all jobs passed or something failed. They show all related jobs for a single commit and the net result of each stage of your pipeline. You can quickly see what failed and fix it.

The pipeline mini graph always group jobs by stage, and display throughout GitLab when displaying pipeline or commit details.



Stages in pipeline mini graphs are expandable. Hover your mouse over each stage to see the name and status, and select a stage to expand its jobs list.

Downstream pipeline graphs

When a pipeline contains a job that triggers a downstream pipeline, you can see the downstream pipeline in the pipeline details view and mini graphs.

In the pipeline details view, a card displays for every triggered downstream pipeline on the right of the pipeline graph. Hover over a card to see which job triggered the downstream pipeline. Select a card to display the downstream pipeline to the right of the pipeline graph.

In the pipeline mini graph, the status of every triggered downstream pipeline displays as additional status icons to the right of the mini graph. Select a downstream pipeline status icon to go to the detail page of that downstream pipeline.

Pipeline success and duration charts

Pipeline analytics are available on the **CI/CD Analytics** page.

Pipeline badges

Pipeline status and test coverage report badges are available and configurable for each project. For information on adding pipeline badges to projects, see Pipeline badges.

Pipelines API

GitLab provides API endpoints to:

- Perform basic functions. For more information, see Pipelines API.
- Maintain pipeline schedules. For more information, see Pipeline schedules API.
- Trigger pipeline runs. For more information, see:
  - Triggering pipelines through the API.
  - Pipeline trigger tokens API.

Ref specs for runners

When a runner picks a pipeline job, GitLab provides that job's metadata. This includes the Git refspecs, which indicate which ref (such as branch or tag) and commit (SHA1) are checked out from your project repository.

This table lists the refspecs injected for each pipeline type:

Pipeline type	Refspecs
pipeline for branches	`+<sha>:refs/pipelines/<id>` and `+refs/heads/<name>:refs/remotes/origin/<name>`
pipeline for tags	`+<sha>:refs/pipelines/<id>` and `+refs/tags/<name>:refs/tags/<name>`

merge request pipeline	`+refs/pipelines/<id>:refs/pipelines/<id>`
pipeline for workload refs	`+refs/pipelines/<id>:refs/pipelines/<id>`

The refs `refs/heads/<name>` and `refs/tags/<name>` exist in your project repository. GitLab generates the special ref `refs/pipelines/<id>` during a running pipeline job. This ref can be created even after the associated branch or tag has been

BY-SA